



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab Experiment – 07

Breadth-First Search (BFS)

Aim

To implement the Breadth-First Search (BFS) algorithm for graph traversal.

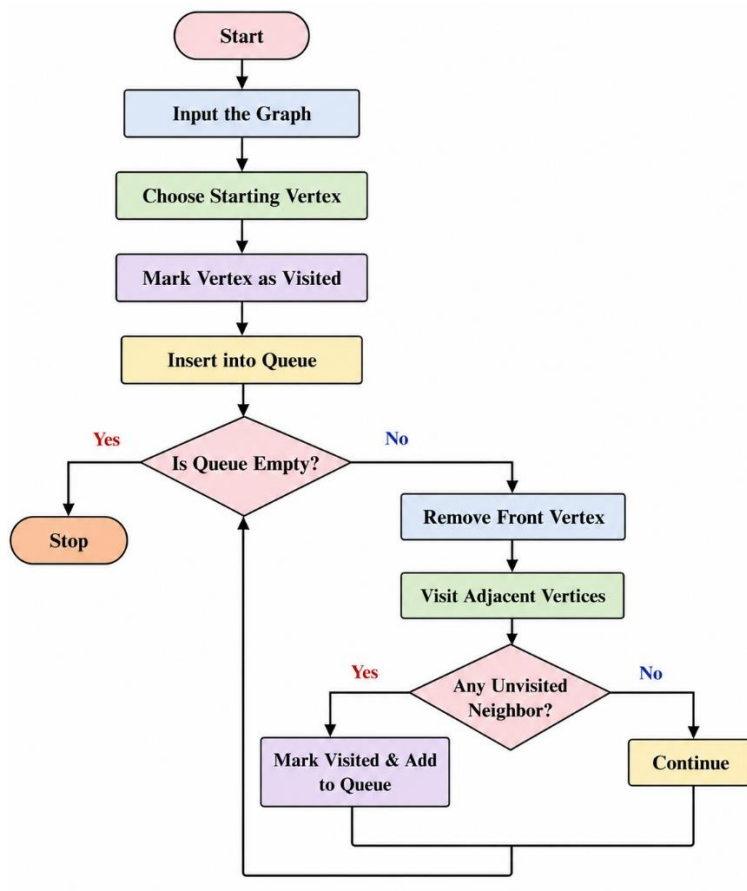
Objective

To traverse all the vertices of a graph level by level using the Breadth-First Search (BFS) algorithm.

Algorithm

1. Start from the source (starting) vertex.
2. Mark the starting vertex as visited.
3. Insert the starting vertex into a queue.
4. Remove the front vertex from the queue.
5. Visit all its unvisited adjacent vertices.
6. Mark each adjacent vertex as visited and insert it into the queue.
7. Repeat Steps 4–6 until the queue becomes empty.
8. Stop.

FlowChart



Code:

```
from collections import deque
```

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': ['F'],  
    'F': []  
}
```

```
visited = set()
```

```
queue = deque()
```

```
start = 'A'
```

```
visited.add(start)
```

```
queue.append(start)
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```
print("BFS Traversal:")
```

```
while queue:
```

```
    vertex = queue.popleft()
```

```
    print(vertex, end=" ")
```

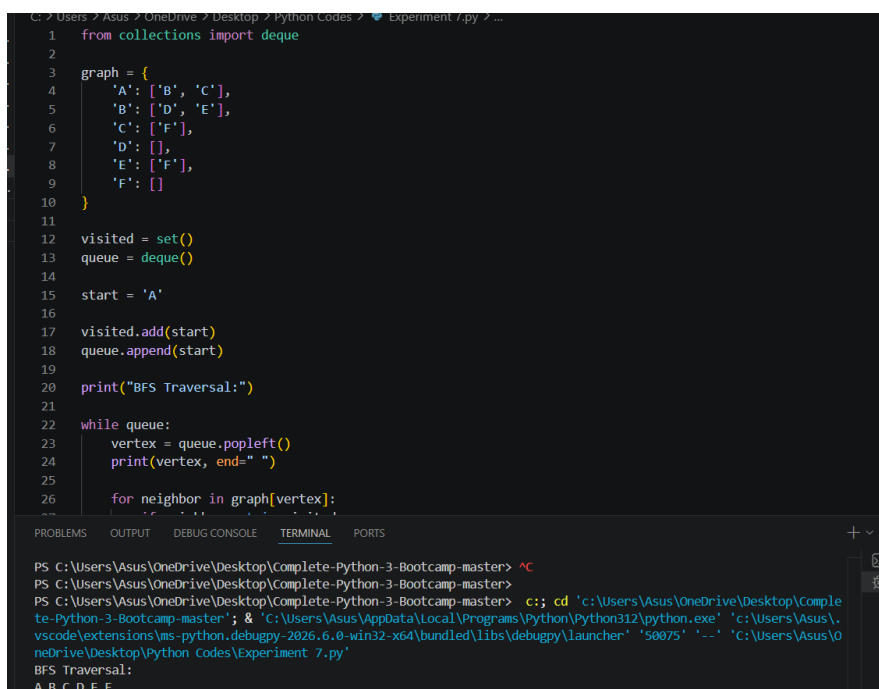
```
    for neighbor in graph[vertex]:
```

```
        if neighbor not in visited:
```

```
            visited.add(neighbor)
```

```
            queue.append(neighbor)
```

Output



```
C:\Users\Asus\OneDrive\Desktop\Python Codes> Experiment 7.py
1 from collections import deque
2
3 graph = {
4     'A': ['B', 'C'],
5     'B': ['D', 'E'],
6     'C': ['F'],
7     'D': [],
8     'E': ['F'],
9     'F': []
10 }
11
12 visited = set()
13 queue = deque()
14
15 start = 'A'
16
17 visited.add(start)
18 queue.append(start)
19
20 print("BFS Traversal:")
21
22 while queue:
23     vertex = queue.popleft()
24     print(vertex, end=" ")
25
26     for neighbor in graph[vertex]:
27         if neighbor not in visited:
28             visited.add(neighbor)
29             queue.append(neighbor)
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```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\Asus\OneDrive\Desktop\Complete-Python-3-Bootcamp-master> ^C
PS C:\Users\Asus\OneDrive\Desktop\Complete-Python-3-Bootcamp-master>
PS C:\Users\Asus\OneDrive\Desktop\Complete-Python-3-Bootcamp-master> c;; cd 'c:\Users\Asus\OneDrive\Desktop\Comple
te-Python-3-Bootcamp-master'; & 'C:\Users\Asus\AppData\Local\Programs\Python\Python312\python.exe' 'c:\Users\Asus\
vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '50075' '--' 'C:\Users\Asus\O
neDrive\Desktop\Python Codes\Experiment 7.py'
BFS Traversal:
A B C D E F
```

Result

The Breadth-First Search (BFS) algorithm was implemented successfully, and all the vertices of the graph were traversed level by level.

Conclusion

The BFS algorithm visits graph nodes in level order using a queue. It guarantees that all reachable vertices are visited systematically and is useful for finding the shortest path in an unweighted graph.